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Rietveld analysis of structure transformation between perovskite-type and B-type rare earth structures

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La*Ln*O_3 (*Ln* = Dy, Ho, Y, Er, and Yb) and La(*Ln*, *Ln*')O_3 (*Ln*, *Ln*' = Dy, Ho, Er, and Yb) systems were synthesized by solid state reaction method and characterized by X-ray diffraction and Rietveld analysis. In order to investigate the size effect of *Ln* site ion in ABO_3-type compound, the phase relationship in the La*Ln*O_3 system. When *Ln* = Er or Yb which has smaller ionic radius than that of Y^{3+} (0.900 Å), the La*Ln*O_3 showed an orthorhombic perovskite-type structure, while when *Ln* = Dy or Ho which has larger ionic radius than that of Y^{3+}, it showed a monoclinic B-type rare earth structure. Next, the solid solution system of $LaHo_xYb_{1-x}O_3$ was investigated in order to clarify the crystallochemical factor affecting the structural transformation. The XRD experiments revealed that the samples with x = 0.90 ($r_{av.}$=0.8977 Å) showed the orthorhombic perovskite-type structure, changed to the mixed phases of monoclinic B-type rare earth, and orthorhombic perovskite-type structures with increasing x, and then the samples with x ≧ 0.95 ($r_{av.}$=0.8994 Å) showed the monoclinic B-type rare earth structures, where $r_{av.}$ represents the average ionic radii of *Ln* and *Ln*'. Therefore, in order to search into process of phase transformation from orthorhombic perovskite-type structure to monoclinic B-type rare earth structure, $LaHo_xYb_{1-x}O_3$ (0.90≦x≦0.95) system was performed Rietveld analysis.

Key words: perovskite-type structure, B-type rare earth structure, ionic radius, Rietveld analysis

1. Introduction

Inasmuch as have properties such as mechanical strength, chemical stability, and high conductivity, which are affect by introduction of defection and size of the space, and etc., fluorite-type related structure (cubic fluorite-type (Fig.1a), cubic pyrochlore-type (Fig.1b) cubic C-type rare earth (Fig.1c), monoclinic B-type rare earth (Fig.1d), and hexagonal A-type rare earth structures (Fig.1e)) and perovskite-type related structure (perovskite-type (Fig.1f), brownmillerite-type (Fig.1g), and rhenium oxide-type structures (Fig.1h)) are paid much attention as electrolytes of solid oxide fuel cell due to their oxide ion conductivities. Among La*Ln*O_3 (Ln: Y or rare earth), $LaYO_3$ is an interesting compound from the viewpoint of crystal chemistry, because firing at 1400℃ showed a low temperature form with orthorhombic perovskite-type structure (Fig.1e), while the firing at 1600℃ transformed to the high temperature form with a monoclinic B-type rare earth structure (Fig. 1c) [1, 2].

The reason why the structure transformation occurs dependent on the firing temperature has not been apparent. In the present study, in order to investigate the crystallochemical factor effecting on the structure transformation, we focused attention on La*Ln*O_3 and La(*Ln*, *Ln*')O_3.

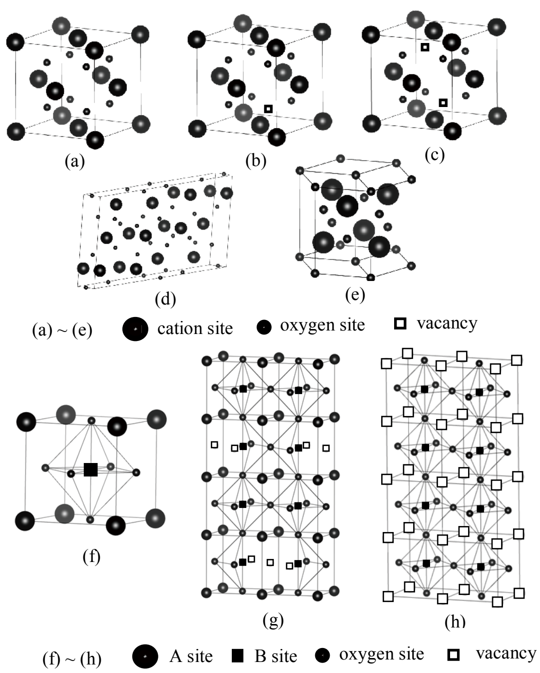

Fig.1 Schematic representation of fluorite-type rerated structure and perovskite-type structure. ((a) Fluorite-type (b) Pyrochlore-type (c) C-type rare earth (d) B-type rare earth (e) A-type rare earth (f) Perovskite-type (g) Brownmillerite-type (h) Rhenium oxide-type)

2. Experimental

Powder samples of the present systems, La*Ln*O_3 (*Ln* = Dy, Ho, Y, Er, Yb), La(*Ln*, *Ln*')O_3 (*Ln*, *Ln*' = Dy, Ho, Er, Yb) were synthesized by means of a solid state reaction method, using La_2O_3 (99.99 %, High Purity Chemicals), Dy_2O_3, Ho_2O_3,Y_2O_3, Er_2O_3, and Yb_2O_3 (99.9 %, High Purity Chemicals) as starting materials. The weighed powders were wet ball-milled for 24 h, using a milling pot made of synthetic resin and resin-coated balls, and ethanol as dispersion reagent. After drying, the powder mixtures were calcined at 1000℃ for 10 h in air. After sieving powders under 53 μm in mesh size, the powder samples were molded under the pressure of 5 MPa and subjected to rubber press at 200 MPa. The compacts thus obtained were sintered at 1400℃ and 1600℃ for 10 h in air, where the heating and cooling rates were 5 ℃/min.

The powdered samples were characterized by means of X-ray diffraction (XRD) (model: MultiFlex, Rigaku) with monochromated CuKα radiation at room temperature. The lattice constant was determined from the XRD peaks by least squares method. In order to carry out the Rietveld analysis, XRD data was collected in the 2θ scanning range from 10° to 140° with a step interval of 0.02° at room temperature. The data was stimulated up to 20000 counts for the strongest diffraction peak. The Rietveld analysis was carried out using the RIETAN-FP application software [3].

3. Results and Discussion

Fig. 2 shows XRD patterns of $LaYO_3$ prepared by firing at 1400℃ and 1600℃. It was confirmed that $LaYO_3$ showed a low temperature form of orthorhombic perovskite-type structure by firing at 1400℃, while it showed a high temperature form of monoclinic B-type rare earth structure by firing at 1600℃ [1]. Then, the samples sintered at 1600℃ were annealed at 1400℃ for 10 h or 1200℃ for 10 h, as shown in Fig. 3. The annealed sample was transformed from the monoclinic B-type rare earth structure to the orthorhombic perovskite-type structure, suggesting the reversible transformation of crystal structure.

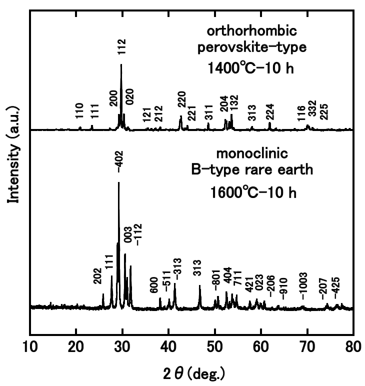

Fig.2 XRD patterns of $LaYO_3$ prepared by firing at 1400℃ and 1600℃.

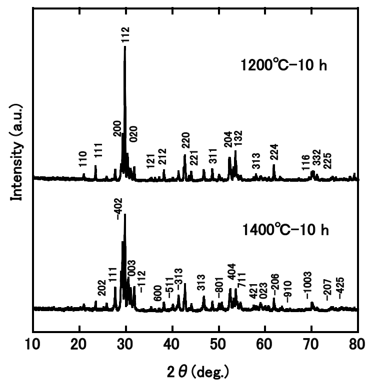

Fig.3 XRD patterns of $LaYO_3$ annealed at 1400℃ and 1200℃ for 10 h after the 1600℃ firing.

Fig.4 shows Rietveld analysis of XRD data of $LaYO_3$ prepared by firing at 1400℃. The sample was analyzed, assuming the single phase of orthorhombic perovskite-type structure (S.G. : *Pnma*). The result of Rietveld analysis showed R_{wp}=14.8%, and S=2.43 suggesting insufficient fitting. Therefore, the sample was again analyzed, assuming two phases of orthorhombic perovskite-type and monoclinic B-type rare earth (S.G. : *C*2/*m*) structures. The result showed R_{wp}=10.5% and S=1.73, which were the lower values comparing with the case of single phase. Therefore, it was found that $LaYO_3$ prepared at 1400 ℃ is composed of 96.4% orthorhombic perovskite-type structure and 3.6% monoclinic B-type rare earth structure (Fig.5).

In order to investigate the size effect of *Ln* ion, the phase relationship in the La*Ln*O_3 (*Ln* = Dy, Ho, Y, Er, Yb) systems [4], where *Ln* has a larger or smaller ionic radius than that of Y^{3+}, was

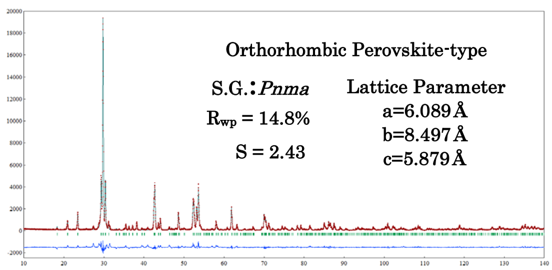

Fig.4 Rietveld analysis of $LaYO_3$ prepared by firing at 1400℃. (single phase)

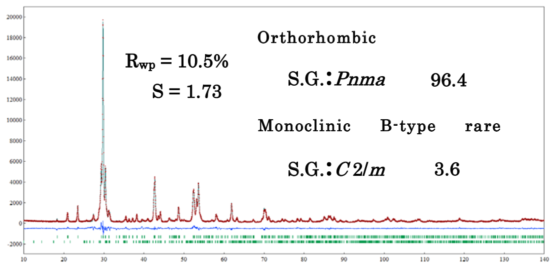

Fig.5 Rietveld analysis of $LaYO_3$ prepared by firing at 1400℃. (two phases)

investigated. The ionic radii of Ln^{3+} are summarized in Table I [5].

In the La*Ln*O3 systems, when *Ln* = Er (0.890Å) or Yb (0868Å) which has the smaller the ionic radius than that of Y^{3+}, the sample showed the orthorhombic perovskite-type structure regardless 1400℃ or 1600℃ firing, while when *Ln* = Dy, (0.912Å) or Ho (0.901Å) which has larger the ionic radius than that of Y^{3+}, it showed the monoclinic B-type rare earth structure regardless the firing temperatures, as shown in Fig. 6 and 7. These facts suggest that the crystal structure of La*Ln*O3 system strongly dependents on the ionic size of Ln^{3+}.

In order to investigate in detail the size effect of Ln^{3+} affecting the crystal structure, the solid solution systems of La(*Ln*, *Ln*')O3 (*Ln*, *Ln*' = Dy, Ho, Er, Yb) were investigated, where the composition ratio, *Ln/Ln*', was controlled for the average ionic radius of *Ln* and *Ln*' ($r_{av.}$) to be in the vicinity of the ionic radius of Y^{3+}. Table II shows the compositional dependence of average

Table I Ionic radius (Å) of Ln^{3+}.

Ln^{3+}	Dy^{3+}	Ho^{3+}	Y^{3+}	Er^{3+}	Yb^{3+}
ionic radius	0.912	0.901	0.900	0.890	0.868

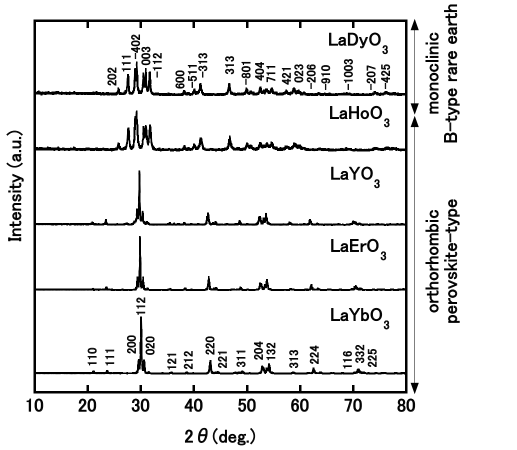

Fig.6 XRD patterns of La*Ln*O3 prepared by firing at 1400℃.

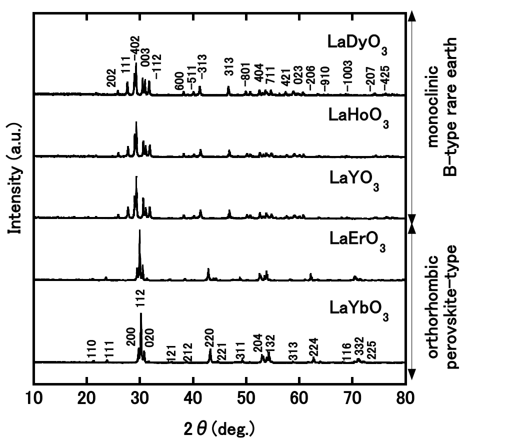

Fig.7 XRD patterns of La*Ln*O3 prepared by firing at 1600℃.

ionic radius in the system with *Ln* = Ho and *Ln*' = Yb.

In the system $LaHo_xYb_{1-x}O_3$ ($0.90 \leqq x \leqq 0.95$), XRD patterns showed mixed phases of the monoclinic B-type rare earth and orthorhombic perovskite-type structures when the samples were sintered at 1400℃ for 10 h. Therefore, the samples were annealed at 1300℃ for 10 h. The samples with x = 0.90 ($r_{av.}$=0.8977Å) showed the orthorhombic perovskite-type structure, and then changed to the mixed phases of the orthorhombic perovskite-type and monoclinic B-type rare earth structures in the composition range, $0.90 < x \leqq 0.95$, as shown in Fig.8. Fig.9 shows XRD patterns of $LaHo_xYb_{1-x}O_3$ system prepared by firing at 1600℃ for 20 h. The samples with x = 0.90 ($r_{av.}$=0.8977 Å) showed the orthorhombic perovskite-type structure by firing at 1600℃ for 20 h, and then changed to the mixed phases of the monoclinic B-type rare earth and orthorhombic perovskite-type structures in the composition range, $0.90 < x < 0.95$, and when x = 0.95 ($r_{av.}$=0.8994 Å), the sample showed the monoclinic B-type rare earth structure.

In order to search into process of phase transformation from orthorhombic perovskite-type structure to monoclinic B-type rare earth structure, samples that $LaHo_xYb_{1-x}O_3$ ($0.90 \leqq x \leqq 0.95$) system prepared by firing at 1600℃ for 20 h was performed Rietveld analysis. The samples with x=0.90 and 0.95 are performed analysis of single phase and $0.90 < x < 0.95$ are two phases model. Phase existence ratio of orthorhombic perovskite-type and monoclinic B-type rare earth structures was shown in table III, respectively. The results obtained by Rietveld analysis were

Table II Average ionic radius (r_{av}/Å) in the $LaHo_xYb_{1-x}O_3$ system.

x	0.95	0.94	0.93	0.92	0.91	0.90
r_{av}	0.8994	0.8990	0.8987	0.8984	0.8980	0.8977

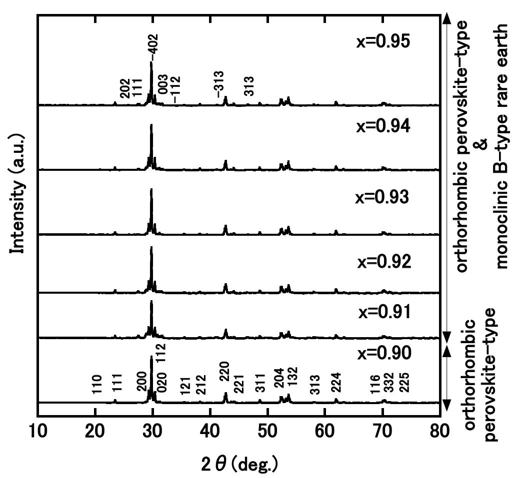

Fig.8 XRD patterns of $LaHo_xYb_{1-x}O_3$ firing 1400℃ and annealed at 1300℃.

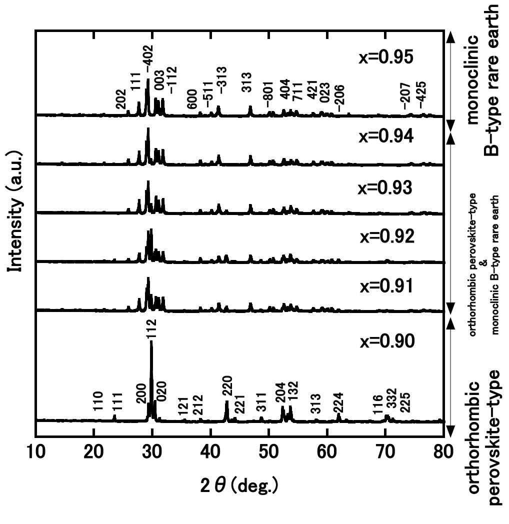

Fig.9 XRD patterns of $LaHo_xYb_{1-x}O_3$ prepared by firing at 1600℃.

plotted phase existence ratio as a function of average ionic radius of (*Ln*, *Ln*') site (Fig.10). With increasing average ionic radius of B site, phase existence ratio was obtained linearly decreasing trend for orthorhombic perovskite-type structure, and it was obtained linearly increasing trend for monoclinic B-type rare earth structure.

Furthermore, $LaEr_xHo_{1-x}O_3$ system was also produced same as average ionic radius of (*Ln*, *Ln*') in $LaHo_xYb_{1-x}O_3$ system, and $LaEr_xHo_{1-x}O_3$ system was performed Rietveld analysis. As a result, when average ionic radius of (*Ln*, *Ln*') which has smaller than r_{av} = 0.8977Å, the sample showed the orthorhombic perovskite-type structure, while when average ionic radius of (*Ln*, *Ln*') which has larger than r_{av} = 0.8994Å, it showed the monoclinic B-type rare earth structure. Therefore, in the La(*Ln*, *Ln*')O_3 systems, controlling for the average ionic radius of (*Ln*, *Ln*'), we could produce the orthorhombic perovskite-type or monoclinic B-type rare earth structures.

Table Ⅲ Phase existence ratio as a function of average radius of (*Ln*, *Ln*') in the $LaHo_xYb_{1-x}O_3$ system.

x	Average ionic radius of (*Ln*, *Ln*') (Å)	Orthorhombic perovskite-type (%)	Monoclinic B-type rare earth (%)
0.95	0.8994	0.0	100
0.94	0.8990	9.8	90.2
0.93	0.8987	27.4	72.6
0.92	0.8984	61.4	38.6
0.91	0.8980	82.2	17.8
0.90	0.8977	100	0.0

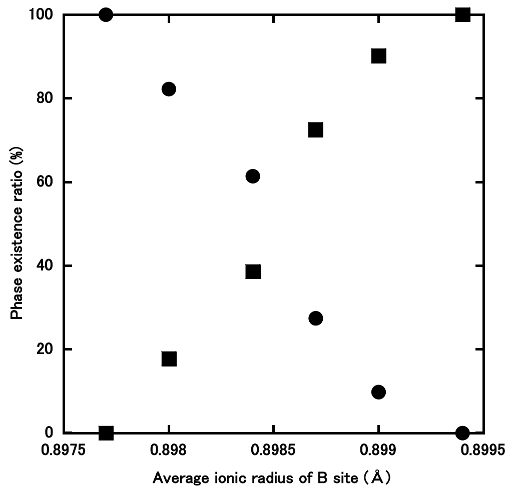

Fig.10 Phase existence ratio as a function of average ionic radius of (*Ln*, *Ln*') in the $LaHo_xYb_{1-x}O_3$ system.

● Orthorhombic perovskite-type structure
■ Monoclinic B-type rare earth structure

4. Conclusion

In the La*Ln*O_3 system, when *Ln* = Er or Yb which has smaller ionic radius than that of Y (0.900 Å), the samples showed an orthorhombic perovskite-type structure, while when *Ln* = Dy or Ho which has larger ionic radius than that of Y, it showed a monoclinic B-type rare earth structure. In the $LaHo_xYb_{1-x}O_3$ ($0.90 \leqq x \leqq 0.95$)system, the samples with x = 0.90 showed the orthorhombic perovskite-type structure, and then changed to the mixed phases of the monoclinic B-type rare earth and orthorhombic perovskite-type structures in the composition range, 0.90<x<0.95, and when x = 0.95, the sample showed the monoclinic B-type rare earth structure by firing at 1600℃ for 20 h. Therefore, the monoclinic B-type rare earth structure was stable when the ionic radius is larger than 0.8994Å, and the orthorhombic perovskite-type structure was stable when the ionic radius is smaller than 0.8977 Å. Consequently, La*Ln*O_3 (*Ln* = Dy, Ho, Y, Er, Yb) and La(*Ln*, *Ln*')O_3 (*Ln*, *Ln*' = Dy, Ho, Er, Yb) systems dependents on the ionic size of Ln^{3+} or (*Ln*, *Ln*').

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